FINAL RESEARCH OUTPUTS



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INTRODUCTION

Cybersecurity education faces the challenge of aligning theoretical knowledge from academic settings with the real-time decision-making required in high-stakes environments. Game-based learning (GBL) has gained interest for boosting engagement, but evidence on its effectiveness for long-term knowledge retention in cyber-risk governance is limited. Addressing this gap is essential for equipping students and professionals with skills to navigate contemporary cybersecurity threats.

This study introduces Breach Point, an innovative card-driven strategy game simulating risk governance principles in a competitive context. Players make strategic decisions on resource allocation, cybersecurity controls, and responses to evolving threats, mirroring industry pressures. The game has evolved from tangible cards to a digital format, enhancing user interaction, clarifying gameplay, and providing a nuanced depiction of budgetary constraints.

This progress report covers eight weeks of research and design, starting with a literature review on GBL and long-term retention. It details methodological advancements, including digital interface improvements and incident card designs, and outlines the evaluation framework for player engagement. The report concludes with a summary of achievements, upcoming deliverables, and a roadmap for final project stages.

In summary, this research highlights Breach Point's potential as an innovative educational tool that combines game mechanics with cybersecurity education, aiming to enhance long-term learning outcomes in the field.

LITERATURE REVIEW

1. Introduction to Game-Based Learning (GBL)

Game-Based Learning (GBL) is an innovative educational approach that utilizes interactive play to engage learners effectively. This method is significant in the educational landscape as it encourages the development of critical thinking, problem-solving, and decision-making skills through real-world scenarios. According to Gee (2003), GBL immerses students in engaging and dynamic environments, enhancing their learning experiences.

The importance of GBL in education is underscored by its ability to improve student engagement and facilitate long-term knowledge retention. Research by Anderson and Dill (2000) indicates that GBL can significantly enhance these aspects by simulating real-world decision-making processes.

2. Popular Games and Their Impact on Learning

Several popular games illustrate the effectiveness of GBL in promoting essential skills:

- Ticket to Ride: This game emphasizes strategic thinking and resource management. Players gain insights into logistics and planning, skills that are applicable in various real-world contexts (Masters of Games, n.d.).
- Wingspan: In this game, players engage in complex decision-making and resource management while learning about biodiversity and environmental systems. Its interactive nature makes the learning process both enjoyable and informative (Board Game Quest, n.d.).
- Catan: This game introduces players to the essential concepts of trade and negotiation within a competitive framework. It provides a deeper understanding of economic principles through strategy and interaction (Wikipedia contributors, 2025).

3. Cybersecurity and GBL in Practice

While GBL has gained ground in various fields, its application in cybersecurity education remains relatively underexplored. Recent studies suggest that simulation games, particularly those focusing on strategy and resource management, can effectively teach concepts such as cyber-risk governance and decision-making under pressure (Chen et al., 2022).

An example of this application is **Breach Point**, a game that integrates principles from strategy-based games like Ticket to Ride and Catan with cybersecurity concepts. In Breach Point, players are tasked with making decisions about cybersecurity assets and resource allocation while defending against evolving cyber

threats. This experience mirrors the management of business or logistics networks found in games like Ticket to Ride.

4. Research Gaps and Your Contribution

Despite the growing interest in GBL, there are notable gaps in the literature concerning the teaching of cybersecurity through games, particularly in the integration of digital formats. Current research has largely overlooked the use of strategy board games for cybersecurity education. This research aims to fill that gap by developing Breach Point, a hybrid game designed to impart foundational knowledge in cyber-risk governance through interactive gameplay.

RESEARCH OUTPUTS

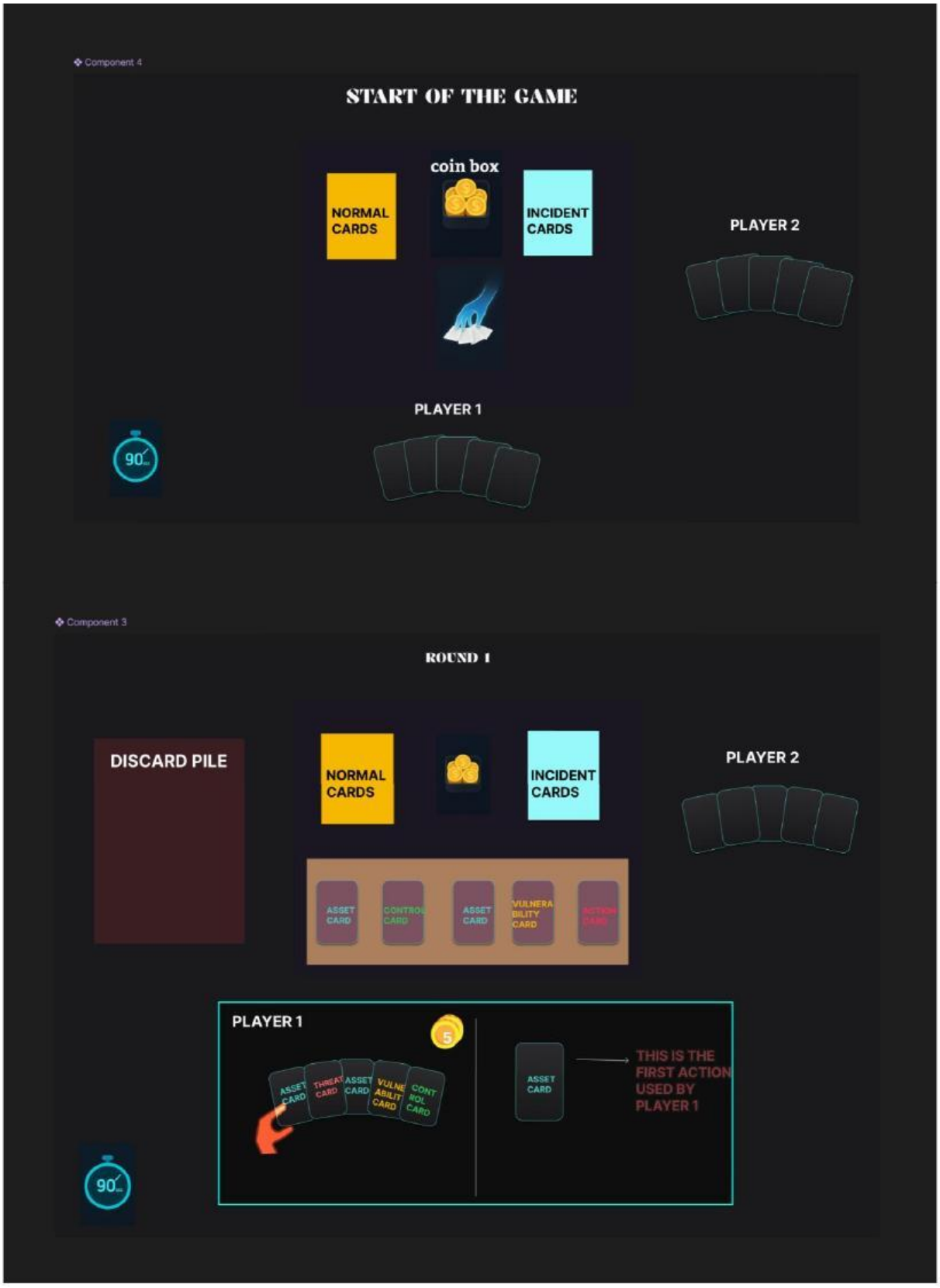
The Breach Point game invites players into a dynamic world of strategic thinking and teamwork. At the start, everyone receives their cards, and key components like the normal cards, incident cards, and the coin box are carefully set up. With a 90-second timer ticking down, there's a sense of urgency that encourages players to make thoughtful decisions in a limited timeframe.

As the game unfolds, players embark on a journey of decision-making, utilizing asset, control, vulnerability, and action cards. Each individual, whether Player 1 or Player 2, takes turns deploying their resources, countering growing threats, and defending against potential cyberattacks. The discard pile becomes a poignant reminder of the choices made, highlighting the ever-changing nature of the game.

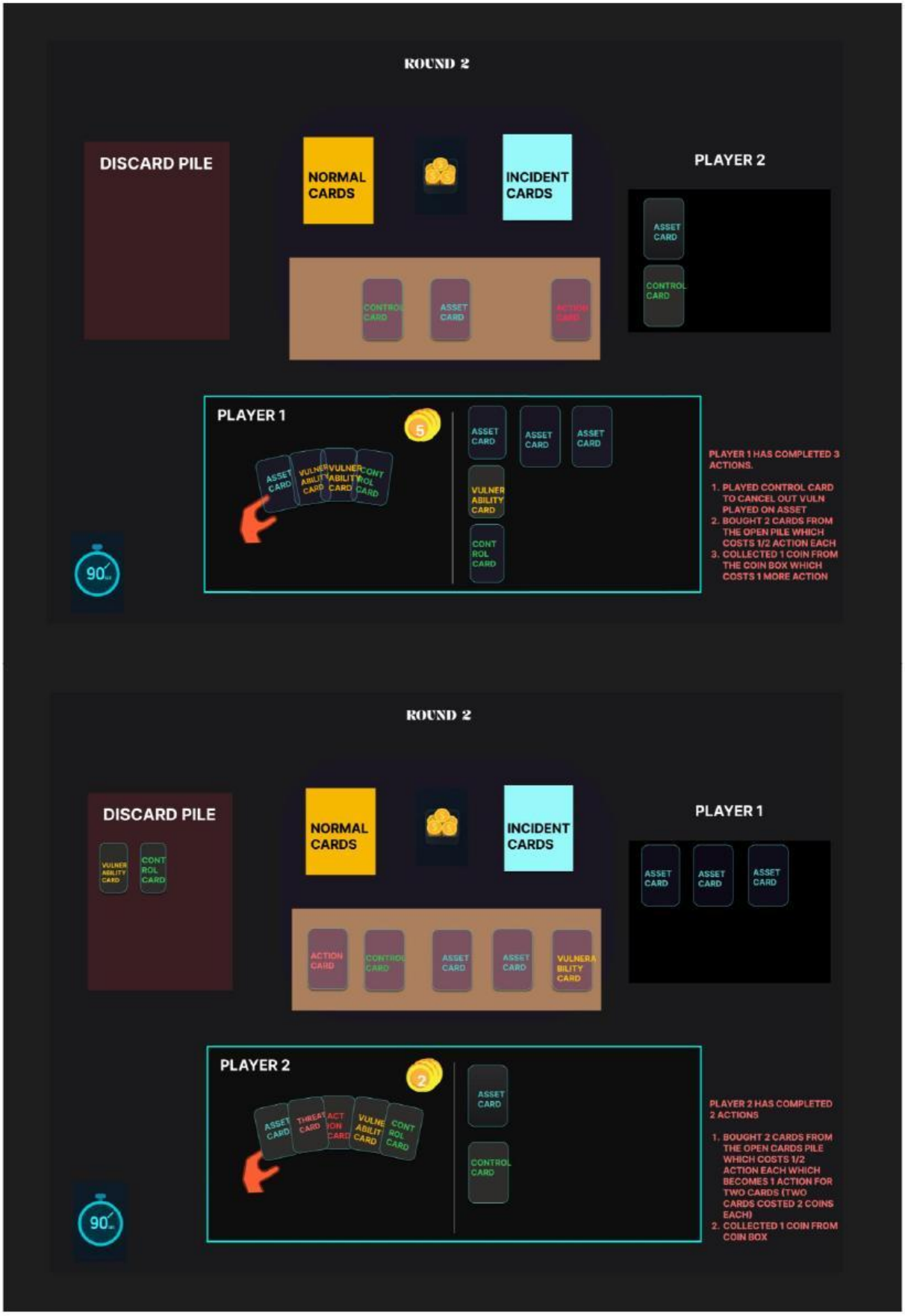
Throughout the rounds, incident cards introduce unexpected scenarios that can catch players off guard, pushing them to adapt and rethink their strategies. The action point system helps track each player's moves, simulating the resource management challenges found in cybersecurity risk governance. As players gather coins from the coin box and complete their actions, the need to adjust strategies becomes essential, reflecting the unpredictable nature of real-world scenarios.

With each passing round, the strategic complexity deepens, requiring players to balance their resources and defend their assets while addressing emerging threats. The game is designed not just for competition but to foster a sense of understanding and growth in decision-making skills, all set against a backdrop that mirrors the realities of managing cybersecurity risks. Its visual clarity and engaging mechanics create a supportive environment that enhances learning and collaboration among players.

OUTPUTS (WORKFLOW ARTIFACT):



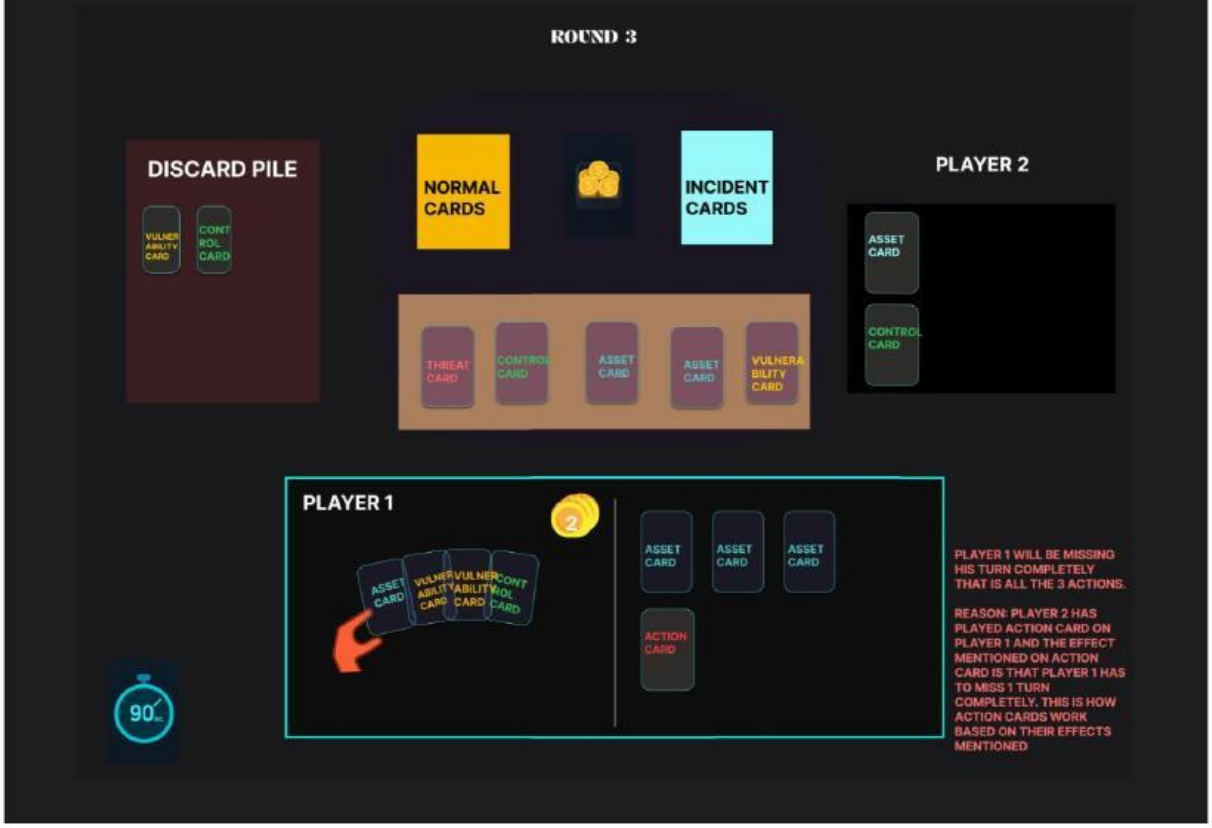

ROUND 1
DISCARD PILE
NORMAL CARDS
INCIDENT CARDS
PLAYER 2
ASSET CARD
CONTROL CARD
ASSET CARD
VULNERABILITY CARD
ACTION CARD
PLAYER 1
5
ASSET CARD
ASSET CARD
ASSET CARD
VULNERABILITY CARD
CONTROL CARD
ASSET CARD
ASSET CARD
ASSET CARD
PLAYER 1 HAS FINISHED ALL OF THE 3 ACTIONS GIVEN BY PLACING ALL THE ASSET CARDS
90
Component 6
ROUND 1
DISCARD PILE
NORMAL CARDS
INCIDENT CARDS
PLAYER 1
ASSET CARD
ASSET CARD
ASSET CARD
VULNERABILITY CARD
ASSET CARD
CONTROL CARD
ASSET CARD
VULNERABILITY CARD
ACTION CARD
PLAYER 2
5
ASSET CARD
THREAT CARD
CONTROL CARD
VULNERABILITY CARD
CONTROL CARD
ASSET CARD
CONTROL CARD
PLAYER 2 HAS COMPLETED THREE ACTIONS.
1. PLACED AN ASSET CARD
2. ATTACKED PLAYER 1 WITH A SUITABLE VULN CARD
3. PROTECTED PLACED ASSET WITH A CONTROL CARD
90


ROUND 2
DISCARD PILE
NORMAL CARDS
INCIDENT CARDS
PLAYER 2
ASSET CARD
CONTROL CARD
CONTROL CARD
ASSET CARD
ACTION CARD
PLAYER 1
5
ASSET CARD
VULNERABILITY CARD
CONTROL CARD
ASSET CARD
ASSET CARD
ASSET CARD
VULNERABILITY CARD
CONTROL CARD
90
PLAYER 1 HAS COMPLETED 3 ACTIONS.
1. PLAYED CONTROL CARD TO CANCEL OUT VULN PLAYED ON ASSET
2. BOUGHT 2 CARDS FROM THE OPEN PILE WHICH COSTS 1/2 ACTION EACH
3. COLLECTED 1 COIN FROM THE COIN BOX WHICH COSTS 1 MORE ACTION
ROUND 2
DISCARD PILE
VULNERABILITY CARD
CONTROL CARD
NORMAL CARDS
INCIDENT CARDS
PLAYER 1
ASSET CARD
ASSET CARD
ASSET CARD
ACTION CARD
CONTROL CARD
ASSET CARD
ASSET CARD
VULNERABILITY CARD
PLAYER 2
2
ASSET CARD
THREAT CARD
VULNERABILITY CARD
CONTROL CARD
ASSET CARD
CONTROL CARD
90
PLAYER 2 HAS COMPLETED 2 ACTIONS
1. BOUGHT 2 CARDS FROM THE OPEN CARDS PILE WHICH COSTS 1/2 ACTION EACH WHICH BECOMES 1 ACTION FOR TWO CARDS (TWO CARDS COSTED 2 COINS EACH)
2. COLLECTED 1 COIN FROM COIN BOX

ACTION CARD



ROUND 3





2ND OUTPUT (INCIDENT CARDS)

The Incident Cards in BreachPoint are designed with care to mirror the complexities and challenges that cybersecurity professionals face in the real world. Each card embodies a unique scenario that encourages players to navigate both the triumphs and trials of cybersecurity, fostering a deeper understanding of the stakes involved in protecting our digital environments.

Take, for instance, the "The Algorithm Asks" card. It invites players to reflect on the very real risks associated with employees clicking on phishing emails. This scenario is a gentle reminder of how crucial human awareness is in cybersecurity, urging players to consider the potential consequences of seemingly innocuous actions and the threats they pose to our systems, such as data breaches or network compromises.

On a more uplifting note, the "Backup Success" card presents a scenario where a player gains a strategic advantage by collecting coins from the Coin Reserve. This highlights the importance of having strong backup and disaster recovery systems in place. It's rewarding to see players rewarded for their foresight in safeguarding their assets, while those who may not have enough coins experience a consequence, which reinforces the need for thoughtful resource management in a supportive way

The "Boardroom Decisions" card places players at a crossroads: to gain an asset right away or forgo their next turn for enhanced security controls. This reflects the often difficult decisions that cybersecurity professionals must balance between short-term gains and long-term safety. It encourages players to thoughtfully weigh their options, mirroring the complexities faced in high-stakes environments where every decision matters.

Then there's the "Zero-Day Exploit" card, which adds an element of uncertainty. Here, players confront the reality of an unpatched vulnerability being misused before a fix can be implemented. The necessary penalty reflects the genuine challenges cybersecurity experts experience when vulnerabilities are exploited, underscoring the urgency of swift action in real situations.

The "Skimmer's Jackpot" card reminds us of the subtle dangers posed by insider threats. It introduces a scenario where one player's hidden exploit impacts others, highlighting the importance of vigilance and proactive measures in creating a safe digital community.

The "Admin Lockout" card brings attention to the potential pitfalls of managing administrative privileges. When this card is activated, it means a player loses an action due to a lockout—an experience that echoes the frustrations professionals feel when access control mechanisms fail.

Lastly, the "Routine Maintenance" card reflects the necessary balance between ongoing upkeep and progress. When drawn, all players must take a moment to slow down and draw one fewer card on their next turn, underlining the reality that regular maintenance, while potentially slowing immediate advancements, is essential for long-term stability.

Collectively, these Incident Cards strive to encapsulate the essence of cybersecurity governance with empathy, encouraging players to make thoughtful decisions on vulnerability management, resource allocation, and strategic planning in uncertain situations. The interplay of these cards creates an engaging experience that encourages players to think critically about risk management, asset protection, and the far-reaching consequences of their choices in the digital world.







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